

GENERAL SPECIFICATION G5.11

Raceways

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G5.11 RACEWAYS

G5.11.1 General

(a) Introduction

- (i) Perform all work; supply, deliver and install all materials and equipment, and installation methods in accordance with, but not limited to, the latest applicable rules, regulations, requirements, national and international codes, as listed in the General Specification G5.2-Standards and Regulations.
- (ii) Conflicts, if any, that may exist between the above reference standards, codes and regulations will be resolved at the discretion of the S.O..

(b) Submittals

- (i) Shop Drawings, Samples and Technical Data: As a minimum, submit the following items:

1) Manufacturer's Literature:

- Rigid galvanised steel conduit;
- Rigid aluminium conduit;
- Stainless Steel Conduit;
- PVC conduit;
- Flexible, non-metallic, liquid-tight conduit;
- Trunking;
- Cable tray/ladder and accessories;
- Mill certificate for SST used in manufacturer of cable tray and cable ladder; and
- Deflection chart.

2) Precast Cast-in-Place Draw Pits and Telecom Manholes:

- Dimensional drawings and descriptive literature;
- Traffic loading calculations; and
- Accessory information.

3) Cable Tray/Ladder Systems:

- Dimensional drawings, calculations, and descriptive information;
- Support span length and kilograms per metre actual and future cable loading at locations, with safety factor used;
- Location and magnitude of maximum simple beam deflection of tray for loading specified;
- Layout drawings and list of accessories;
- Cable tray/cable ladder fill calculations using dimensions of actual cables proposed for use. Allow 25 percent minimum spare space within cable tray/cable ladder when calculating fill;

- Raceways support and raceway loading calculations with safety factor 3 endorsed by structural PE;
- Cable schedule that summarises cables by cable number, including cable sizes, installed in each raceway section; and
 - Raceway section defined as each section of cable tray or cable ladder bounded by junctures and bends; and
 - Raceway section also defined as each unbroken section of conduit, trunking or duct.
- Minimum three (3) samples of cable ladder for the S.O.'s approval.

G5.11.2 Products

(a) Conduit and Tubing

(i) Rigid Galvanised Steel Conduit (RGS)

- 1) Conduit shall conform to IEC 61386 and shall be heavy gauge, screwed at both ends, solid drawn and seamed by welding;
- 2) Conduit shall show no appreciable unevenness and be free from internal burns, fins and the like which may cause damage to cables;
- 3) Material: Internal and external hot-dip galvanised, with chromated protective layer;
- 4) Conduit shall have protection against corrosion according to Class 4 as specified in IEC 61386; and
- 5) Protective coating on conduit and its associated fittings shall be identical.

(ii) Rigid Aluminium Conduit

- 1) Material: Copper-free anodized aluminium alloy.

(iii) Rigid PVC Conduit (UPVC)

- 1) Meet requirements of applicable parts of BS 4607;
- 2) Impact resistant, UV stabilised PVC;
- 3) Heavy-duty type (minimum wall thickness from 2.4 to 5 mm);
- 4) Plain ends with slip-on couplings;
- 5) Conduit Colour: White or Grey unless stated otherwise by S.O.; and
- 6) A protective coat of white enamel paint shall be painted on conduit mounted outside building exposed to weather.

(iv) PVC-Coated Rigid Galvanised Steel Conduit

1) Material

- Meet requirements for rigid galvanised steel conduit as specified above;
- Exterior Finish: PVC coating, 1 mm nominal thickness, bond to metal shall have tensile strength greater than PVC; and
- Interior Finish: Urethane coating, .05 mm nominal thickness.

- 2) Threads: Hot-dipped galvanised and factory coated with urethane; and
- 3) Bendable without damage to either interior or exterior coating.
- (v) Flexible, Non-Metallic, Liquid-Tight Conduit
 - 1) Material: PVC core with fused flexible PVC jacket.
- (b) Fittings
 - (i) Rigid Galvanised Steel Conduit Fittings
 - 1) Meet requirements of IEC 61386, no elbows or tees are permitted;
 - 2) Fittings shall have internal fixing lugs;
 - 3) Materials: Malleable cast iron and be hot-dipped galvanised;
 - 4) Adaptable boxes and covers shall be grey cast iron with external fitting lugs;
 - 5) Fittings and boxes with covers shall have a machined surface around the perimeter mating surface with similar machined surface on the cover;
 - 6) Cover fixing screws shall be brass;
 - 7) The protective coating on boxes, covers, metal partitions and associated conduit shall be identical;
 - 8) Internally screwed flange couplers shall be of malleable cast iron:
 - Dimensions:
 - Overall Length: "D" plus 12 mm nominal;
 - Diameter of Flange: "D" plus 12.7 mm nominal; and
 - Thickness of Flange: 3.175 mm nominal.
 - Couplers shall be provided with a neoprene washer and steel washer.
 - 9) All conduit outlet boxes and junction boxes shall be of malleable iron and standard circular pattern with spout to IEC 61386;
 - 10) Deflection/Expansion Fittings
 - Neoprene sleeve connecting two end couplings capable of allowing a minimum of 15 mm of movement in any direction and a deflection of 30 degrees;
 - Stainless steel bands to fix the neoprene sleeve to the end couplings; and
 - Assemble shall include a tinned copper braid to provide an earth path across the neoprene sleeve.
 - 11) Cable Glands
 - Conform to requirements BS 6121;
 - To form watertight nonslip cable connection to enclosure or conduit;
 - Material: Stainless steel with neoprene bushing to assure watertight seal; and

- Shall be specifically designed for type and size of cable being terminated.
- (ii) Rigid Aluminium Conduit
 - 1) General:
 - Type: Threaded, copper-free cast aluminium. Set screw fittings not permitted; and
 - Finish: Equivalent to that of the conduit.
- (iii) PVC Conduit and Tubing
 - 1) Meet requirements of BS 4607;
 - 2) Type: PVC, slip-on with manufacturer's standard sealing cement; and
 - 3) Colour: White, identical to conduit used, unless stated otherwise by S.O..
- (iv) PVC-Coated Rigid Galvanised Steel Conduit
 - 1) Fittings: Rigid galvanised steel type, PVC-coated by conduit manufacturer;
 - 2) Conduit Bodies: Cast metal hot-dipped galvanised or urethane finish. Cover shall be of same material as conduit body. PVC-coated by conduit manufacturer;
 - 3) Finish: 1 mm PVC exterior, 50 microns urethane interior;
 - 4) Overlapping pressure sealing sleeves; and
 - 5) Conduit Hangers, Attachments and Accessories: PVC-coated.
- (i) Stainless Steel Conduit
 - 1) Conduits shall be fitted at regular intervals of not exceeding 700 mm on horizontal runs and 900 mm for vertical runs;
 - 2) A Junction or adapter boxes shall be provided in all conduit runs exceeding 10 m in length or containing more than two right angle bends to facilitate the draw-in wires; and
 - 3) Junction and adapter boxes shall be made of stainless steel. Outdoor fittings enclosures shall be weather proof stainless steel boxes.
- (ii) Flexible, Non-Metallic, Liquid-Tight Conduit
 - 1) Type: High strength plastic body, complete with lock nut, O-ring, threaded ferrule, sealing ring and compression nut; and
 - 2) Body/compression nut (gland) design to assure high mechanical pull-out strength and watertight seal.
- (iii) Flexible Coupling, Hazardous Locations
 - 1) Approved for use in the atmosphere involved;
 - 2) Rating: Watertight for use in Zone 0, 1 and 2 hazardous areas;
 - 3) Outer bronze braid and an insulating liner; and
 - 4) Conductivity equal to a similar length of rigid metal conduit.

- (c) Stainless Steel Trunking and Fittings
 - (i) Shall be manufactured in compliance with BS 4678: Part 1 and SS 249;
 - (ii) Shall be used only with the written consent of the S.O., where multiple parallel conduit runs occur;
 - (iii) Fabricated from Type 316 stainless steel;
 - (iv) 1.2 mm minimum thickness for up to 100 x 100 mm, and 1.6 mm for larger sizes;
 - (v) The number of cables installed in each trunking shall be such that the space factor of 25% is not exceeded;
 - (vi) Shall be supported at intervals not exceeding 2 m horizontally or 2.5 m vertically;
 - (vii) Where standard length of steel trunking are cut, Contractor shall prior to assembly, treat the cut edge with two coats of cold galvanized compound followed by two coats of polyester paint;
 - (viii) Trunking system shall be at least 150 mm clear of plumbing and mechanical services;
 - (ix) Manufacturer's standard fittings shall be used;
 - (x) Standard flanged couplings shall be used to terminate trunking at apparatus and enclosures. Field cutting and bending of material to form a flange attachment will not be accepted; and
 - (xi) Provide external copper earth bonding link on both sides of all joints. Cross sectional area shall not be less than 4 mm².
- (d) PVC Trunking and Accessories
 - (i) Shall be manufactured in compliance with BS 4678: Part 4 and meet applicable requirements of SS CP 5 and BS 4678;
 - (ii) Mandatory marking code 405/1222 in accordance to marking classification of BS 4678;
 - (iii) PVC trunking shall be of the non-flame propagating type with cover secure;
 - (iv) PVC trunking shall be IP 33 (minimum) with cover secure;
 - (v) Manufacturer's standard accessories shall be used; and
 - (vi) All PVC trunking and accessories shall be standard white colour.
- (e) Perforated Metal Cable Trays and Accessories
 - (i) Manufactured to IEC 61537;
 - (ii) Materials:
 - 1) Epoxy-Coated Galvanised Sheet Steel:
 - Perforated Galvanised Mild Steel Sheet, 1.6 mm Thick (Minimum);
 - Hot-dip galvanised to BS EN ISO 1461 after fabrication; and
 - Finish: minimum 45-micron powdered epoxy coating.

- 2) Perforated Type 316 stainless steel sheet, 1.5 mm thick (minimum).
- (iii) Manufacturers' standard fittings shall be used;
- (iv) Bends shall have an internal radius not less than 200 mm and a straight length of not less than 100 mm at each end;
- (v) Flange Height: 25 mm minimum, 50 mm for widths 100 mm and greater;
- (vi) Fittings and accessories used shall be of manufacturers' standard items and shall be of same material, thickness, and finish as the cable tray;
- (vii) Mushroom-head stainless steel roofing bolts and nuts or round head square necks with spring washers and flange nuts complying with BS 4A 114 shall be used to fix together adjacent sections of cable tray and/or accessories;
- (viii) Cover: Perforated and flanged, minimum 1.2 mm thick;
- (ix) Barrier Strip: Vertical, solid type with horizontal fittings and strip clamps; and
- (x) Supply, delivery and install cable tray with no sharp edges, burrs or weld projections.
- (f) Cable Ladder and Accessories
 - (i) Manufactured to IEC 61537;
 - (ii) Material:
 - 1) Epoxy-Coated Electro-Galvanised Sheet Steel:
 - Hot Rolled Steel to JIS G3131 SPHC, 2 mm Thick Minimum;
 - Hot-dip galvanised to BS EN ISO 3834 after fabrication; and
 - Finish: minimum 45-micron powdered epoxy coating.
 - 2) Type 316 stainless steel, 2 mm thick minimum; and
 - 3) Cable ladder installed shall be equal to or exceed the quality of the sample approved by the S.O..
 - (iii) Welded rungs on 300 mm centres;
 - (iv) Bends shall have an internal radius not less than 300 mm and a straight length of not less than 100 mm at each end;
 - (v) Rail Height: minimum 150 mm;
 - (vi) Fittings and accessories used shall be of manufacturers' standard items and shall be of same material, thickness and finish as the cable ladder;
 - (vii) Mushroom-head stainless steel roofing bolts and nuts or round head square necks with spring washers and flange nuts complying with BS 4A-114 shall be used to fix together adjacent sections of cable ladder and/or accessories;
 - (viii) Supply, delivery and install cable ladder with no sharp edges, burrs or weld projections. Use TIG Fine Fusion weld between rung and side rail. Fittings shall be of the continuous arc type with radius as required for elbows, tees, cross, risers and reducers etc;
 - (ix) Cover: Perforated and flanged, minimum 1.2 mm thickness;

- (x) Barrier Strip: Vertical, solid type, with horizontal fittings and strip clamps,
- (xi) Rail with returned flange;
- (xii) Tray Grounding: Conform to CP 5 and SS 551; and
- (xiii) Warning Signs: 37mm high black lettering on yellow background with legend, "WARNING, NOT TO BE USED AS WALKWAY, LADDER, OR SUPPORT FOR LADDERS OR PERSONNEL."
- (g) Draw Pits and Telecom Manholes
 - (i) Provide draw pits and Telecom manholes as specified on Drawings shall be designed in accordance with local codes; and
 - (ii) Provide Telecom manholes in full compliance with IMDA COPIF requirements.
- (h) Accessories
 - (i) Wraparound Duct Band
 - 1) Material: Heat-shrinkable, cross-linked polyolefin, pre-coated with hot-melt adhesive; and
 - 2) 50 mm width (minimum).
 - (ii) Identification Devices
 - 1) Raceway Tags:
 - Material: Permanent, non-ferrous metal;
 - Shape: Round;
 - Raceway Designation: Pressure stamped, embossed, or engraved; and
 - Tags relying on adhesives or taped-on markers not permitted.
 - (iii) Warning Tape
 - 1) Material: Polyethylene, 0.1 mm with detectable strip;
 - 2) Colour: Red;
 - 3) Width: Minimum 150 mm;
 - 4) Designation: Warning on tape that electric circuit is located below tape; and
 - 5) Identifying Letters: Minimum 25 mm high permanent black lettering imprinted continuously over entire length.
 - (iv) Buried Raceway Marker
 - 1) Material: Sheet bronze, consisting of double ended arrows, straight for straight runs and bent at locations where runs change direction;
 - 2) Designation: Incise to depth of 3 mm, ELECTRIC CABLES, in letters 6 mm high; and
 - 3) Minimum Dimension: 10 mm thick, 250 mm long and 25 mm wide.

- (v) Cable Cover Plate (For Buried Cables)
 - 1) Material: High impact resistant rigid PVC, having angle on both sides to allow the plate to be firmly held to the ground complete with interlocking device between plates;
 - 2) Designation: Marked indelibly ELECTRICAL CABLES, in letters at least 45 mm high and 30 mm wide; and
 - 3) Minimum Dimension: 2 mm thick, 1,000 mm long and 150 mm wide.
- (vi) Duct Bank Spacers
 - 1) Type: Non-metallic, interlocking, for multiple conduit sizes; and
 - 2) Suitable for all types of conduit.

G5.11.3 Execution

- (a) General
 - (i) Conduit sizes shown are based on the use of copper conductors;
 - (ii) Proposed conduit routes shall be part of the co-ordinated layout drawings to be submitted to the S.O. for approval prior to installation;
 - (iii) Suitable battens and blocks shall be provided and fixed in the shuttering before concrete is poured to form chases in the concrete work where applicable. Work shall be closely liaised with all contractors and shall not interfere with or delay the progress of the building work;
 - (iv) Prior permission shall be obtained from the S.O. for coring, cutting and chasing in/on reinforced concrete structures. Any unauthorised coring, cutting or chasing shall be liable for penalty to be determined by the S.O.;
 - (v) Crushed or deformed raceways not permitted;
 - (vi) Maintain raceway entirely free of obstructions and moisture;
 - (vii) Immediately after installation, plug or cap raceway ends with watertight and dust-tight seals until time for pulling in conductors;
 - (viii) All outlet and junction boxes shall be fitted with a protective plate and countersunk screws prior to concreting and plastering. No re-threading shall be allowed;
 - (ix) All conduit entries to adaptable boxes, outlet boxes and switchgear shall be made with couplings and hexagon male bush;
 - (x) Group raceways installed in same area;
 - (xi) Follow structural surface contours when installing exposed raceways. Avoid obstruction of passageways;
 - (xii) Run exposed raceways parallel or perpendicular to walls, structural members, or intersections of vertical planes;
 - (xiii) Block Walls: Do not install raceways in same horizontal course with reinforcing steel;
 - (xiv) Install watertight fittings in outdoor, underground or wet locations;

- (xv) Paint threads and cut ends, before assembly of fittings and galvanised conduit, in exposed or damp locations with zinc-rich paint or liquid galvanising compound;
 - (xvi) Metal conduit to be cut square, reamed, burrs removed, and cleaned before installation of conductors, wires or cables;
 - (xvii) Conduits shall be threaded to correct length to ensure a tight, full length connection in tapped spouts. Excessive length of thread will not be permitted;
 - (xviii) Where conduits are set in a column or beam, each end shall be terminated into a coupling and threaded metal plug to permit subsequent joining;
 - (xix) Do not install raceways in concrete equipment pads, foundations, or beams;
 - (xx) Horizontal raceways installed under floor slabs shall lie completely under slab, with no part embedded within slab;
 - (xxi) Install concealed, embedded and buried raceways so that they emerge at right angles to surface and have no curved portion exposed;
 - (xxii) Conduits shall be installed to be self-draining incorporating as necessary approved means of drainage;
 - (xxiii) All metal conduits shall be electrically continuous throughout and shall be efficiently earthed with copper wire or tape and screw on bushings or clips; and
 - (xxiv) Install conduits for telecom/telephone service provider (PSTN) cables in strict conformance with the requirements of IMDA.
- (b) Installation in Cast-In-Place Structural Concrete
- (i) Minimum cover 50 mm, including all fittings;
 - (ii) Conduit placement shall not require changes in reinforcing steel location or configuration;
 - (iii) Securely fix conduits to upper and lower layers of reinforcing steel to prevent displacement during pouring;
 - (iv) Outlet boxes shall be installed so they will be flush with the final plastering, packing ring or extension ring shall be installed where necessary;
 - (v) Install deflection/expansion fittings in all conduits where they cross an expansion;
 - (vi) Slabs and Walls
 - 1) Standard size of conduit not to exceed one-fourth of the slab or wall thickness;
 - 2) Install within middle third of slab or wall;
 - 3) Separate conduit not less than 150 mm centre-to-centre, unless otherwise shown on the Drawings;
 - 4) Maintain widest spacing possible where conduits converge at panels and junction boxes;
 - 5) Securely support conduits at boxes to prevent dislocation during concrete pouring;

- 6) Cross conduit at an angle greater than 45 degrees, with minimum separation of 25 mm;
- 7) Separate conduit by a minimum six times the outside dimension of expansion/deflection fittings at expansion joints; and
- 8) Conduit shall not be installed below the maximum water surface elevation in walls of water-holding structures.
- (vii) Columns and Beams
 - 1) Standard size of conduit not to exceed one-fourth of beam thickness; and
 - 2) Conduit cross-sectional area not to exceed 4 percent of beam or column cross section.
- (c) Conduit Application
 - (i) Diameter: Minimum 25 mm;
 - (ii) Exterior, Exposed: Rigid aluminium;
 - (iii) Interior, Exposed: Rigid aluminium;
 - (iv) Interior, concealed (Not Embedded in Concrete): Rigid galvanised steel;
 - (v) Aboveground, Embedded in Concrete Walls, Ceilings or Floors: Rigid galvanised steel;
 - (vi) Direct Earth Burial: Not allowed; must be concrete encased;
 - (vii) Concrete-Encased Raceways: PVC;
 - (viii) Under Slabs-On-Grade: PVC, concrete encased;
 - (ix) Corrosive Areas: PVC where physically protected from damage; and
 - (x) Hazardous Gas Areas: Rigid galvanised steel.
- (d) Connections
 - (i) For motors, wall- or ceiling-mounted fans and unit heaters, dry type transformers, electrically operated valves, instrumentation and other equipment supplied by a conduit system where flexible connection is required to minimise vibration:
 - 1) Wet or Corrosive Areas: Flexible, non-metallic liquid-tight;
 - 2) Hazardous Areas: Flexible coupling suitable for Zone 0, 1 and 2 locations; and
 - 3) Length: 300 mm minimum, 1,000 mm maximum, sufficient to allow movement or adjustment of equipment.
 - (ii) Lighting Fixtures in Dry Areas: Unprotected cable or trunking; and
 - (iii) Spanner or purpose made tools shall be used to tighten hexagon bushes. Pliers and toothed wrenches shall not be used.
- (e) Penetrations
 - (i) Make at right angles, unless otherwise shown on the Drawings;

- (ii) Notching or penetration of structural members, including footings and beams, not permitted;
- (iii) Sealing at Walls, Floors and Ceilings
 - 1) Firestop penetrations in fire-rated walls, floors and ceilings to maintain fire-resistance rating as specified in the general specifications; and
 - 2) Seal penetrations and conduits between hazardous and non-hazardous with approved sealing material or devices to provide a gastight seal.
- (iv) Apply single layer of wrap-around duct band to all metallic conduit protruding through concrete floor slabs to a point 50 mm above and 50 mm below concrete surface;
- (v) Concrete Walls, Floors or Ceilings (Aboveground): Provide non-shrink grout dry-pack or use watertight seal device;
- (vi) Entering Structures
 - 1) General: Seal raceway at the first box or outlet with oakum or expandable plastic compound to prevent the entrance of gases or liquids from one area to another.
 - 2) Concrete Roof or Membrane Waterproofed Wall or Floor:
 - Provide a watertight seal;
 - Without Concrete Encasement: Install watertight entrance seal device on each side;
 - With Concrete Encasement: Install watertight entrance seal device on the accessible side;
 - Securely anchor malleable iron body of watertight entrance seal device into construction with one or more integral flanges; and
 - Secure membrane waterproofing to watertight entrance seal device in a permanent, watertight manner.
 - 3) Heating, Ventilating, and Air Conditioning Equipment:
 - Penetrate equipment in area established by manufacturer;
 - Terminate conduit with flexible non-conduit at junction box attached to exterior surface of equipment prior to penetrating equipment; and
 - Seal penetration with Type 5 sealant.
 - 4) Corrosive-Sensitive Areas:
 - Seal all conduit passing through chlorine and ammonia room walls;
 - Seal conduit entering equipment panels and field panels containing electronic equipment;
 - Seal penetration with Type 5 sealant; and
 - Existing or Precast Wall: Core drill wall and install a watertight entrance seal device.
 - 5) Draw Pits:

- Metallic Raceways: Provide insulated earthing bushings; and
 - Non-Metallic Raceways: Provide bell ends flush with wall.
- (f) Support
- (i) Support conduit from structural members only, at intervals not exceeding 1.2 metres for conduit up to and including 25 mm and 1.8 metres for sizes 32 mm and larger. Do not support from piping, pipe supports or other raceways;
 - (ii) Multiple Adjacent Raceways: Provide ceiling trapeze;
 - (iii) Spacer bars shall be used for securing conduits on surfaces other than structural steel members;
 - (iv) Support with approved clamps on structural members;
 - (v) Conduits shall be supported not less than 150 mm and 300 mm respectively from each surface box or switch and shall be one per conduit to each box or switch;
 - (vi) Application/Type of Conduit Strap
 - 1) Aluminium Conduit: Aluminium or stainless steel;
 - 2) Rigid steel malleable iron;
 - 3) PVC-Coated Rigid Steel Conduit: PVC-coated metal; and
 - 4) Non-Metallic Conduit: Non-metallic or PVC-coated metal.
 - (vii) Attach wall brackets, strap hangers or ceiling trapeze as follows:
 - 1) Hollow Masonry Units: Toggle bolts;
 - 2) Concrete or Brick: Expansion shields or threaded studs driven in by powder charge, with lock washers and nuts;
 - 3) Steelwork: Machine screws; and
 - 4) Type of Hardware:
 - All Areas: Stainless steel SS 316.
 - (viii) Nails or wooden plugs inserted in concrete or masonry for attaching raceway not permitted. Do not weld raceways or pipe straps to steel structures. Do not use wire in lieu of straps or hangers.
- (g) Bends
- (i) Install concealed raceways with a minimum of bends in the shortest practical distance.
 - (ii) Make bends and offsets of longest practical radius. Bends in conduits and ducts being installed for fibre optic cables shall be not less than 20 times cable diameter, 400 mm minimum.
 - (iii) Install with symmetrical bends or cast metal fittings.
 - (iv) Conduit runs shall not contain more than the equivalent of three 90-degree bends between pull points.

- (v) Avoid field-made bends and offsets, but where necessary, make with acceptable hickey or bending machine. Do not heat metal raceways to facilitate bending.
- (vi) Make bends in parallel or banked runs from same centre or centreline with same radius so that bends are parallel.
- (vii) Factory elbows may be installed in parallel or banked raceways if there is change in plane of run, and raceways are same size.
- (viii) PVC Conduit
 - 1) Bends 30-Degree and Larger: Provide factory-made elbows;
 - 2) 90-Degree Bends: Provide rigid steel elbows; and
 - 3) Use manufacturer's recommended method for forming smaller bends.
- (ix) Flexible Conduit: Do not make bends that exceed allowable conductor bending radius of cable to be installed or that significantly restricts conduit flexibility.
- (h) Deflection/Expansion Fittings
 - (i) Provide on all raceways at all structural expansion joints, and in long tangential runs;
 - (ii) Install in accordance with manufacturer's instructions; and
 - (iii) Deflection shall be in accordance with IEC 61537.
- (i) PVC Conduit
 - (i) Solvent Welding
 - 1) Provide manufacturer recommended solvent; apply to all joints; and
 - 2) Install such that joint is watertight.
 - (ii) Adapters
 - 1) PVC to Metallic Fittings: PVC terminal type; and
 - 2) PVC to Rigid Metal Conduit: PVC female adapter.
 - (iii) Belled-End Conduit: Bevel the unbelled end of the joint prior to joining.
 - (iv) PVC conduit installed underground shall be protected by a 100 mm deep concrete slab located within 200 mm of the top of the conduit and extending a minimum of 100 mm each side of the conduits if not installed as part of a concrete encased duct bank.
- (j) PVC-Coated Rigid Steel Conduit
 - (i) Install in accordance with manufacturer's instructions; and
 - (ii) All tools and equipment used in the cutting, bending, threading and installation of PVC-coated rigid steel conduit shall be designed to limit damage to the PVC coating.
 - (iii) Provide PVC boot to cover all exposed threading.
- (k) Cable Trunking
 - (i) Install in accordance with manufacturer's instructions;

- (ii) Locate with cover on accessible face of trunking, unless otherwise shown on the Drawings; and
 - (iii) Applications
 - 1) Metal trunking in indoor dry, indoor wet and outdoor locations; and
 - 2) Non-metallic trunking in corrosive locations.
- (l) Cable Trays and Cable Ladders
 - (i) Install in accordance with manufacturer's installation instructions;
 - (ii) Provide accessories as necessary for a complete system;
 - (iii) Install such that joints are not made at support brackets;
 - (iv) Install horizontal section support brackets between support point and quarter point of tray or ladder span;
 - (v) Provide ceiling trapeze for all horizontal cable tray and cable ladder;
 - (vi) Install support within 500 mm on each side of expansion joints and within 500 mm of fitting extremity;
 - (vii) Install horizontal tray and ladder level, plumb, straight, and true to line or grade within a tolerance of 3 mm in 5 metres and within a cumulative maximum of 15 mm;
 - (viii) Install vertical tray and ladder plumb within a tolerance of 3 mm in 5 metres;
 - (ix) Install without exposed raw edges;
 - (x) Maintain 300 mm vertical separation between multi-tiered trays and ladders having a common support and at all crossover locations; and
 - (xi) Provide earthing jumper at each joint and adjustable connection to make entire tray or ladder electrically continuous.
- (m) Termination at Enclosures
 - (i) Cast Metal Enclosure: Provide manufacturer's pre-moulded insulating sleeve inside metallic conduit terminating in threaded hubs;
 - (ii) Non-Metallic, Cabinets and Enclosures: Terminate conduit in threaded conduit hubs, maintaining enclosure integrity;
 - (iii) Sheet Metal Boxes, Cabinets, and Enclosures:
 - 1) Rigid Galvanised or Aluminium Conduit:
 - Provide one lock nut each on inside and outside of enclosure;
 - Install earthing bushing;
 - Provide jumper from earthing bushing to equipment earth bus or earth pad; if neither earth bus nor pad exists, connect jumper to bolt attached to metal enclosure;
 - Install insulated bushing on ends of conduit where earthing is not required;
 - Provide insulated throat when conduit terminates in sheet metal boxes having threaded hubs; and

- Utilise threaded hubs on outside of watertight and dust-tight enclosures.
- 2) Flexible, Non-Metallic Conduit: Provide non-metallic, liquid-tight strain relief connectors;
- 3) PVC-Coated Rigid Galvanised Steel Conduit: Provide PVC-coated, liquid-tight, metallic connector; and
- 4) PVC Conduit: Provide PVC terminal adapter with lock nut.
- (iv) Motor Control Centre, Switchboard, Switchgear and Free-Standing Enclosures:
 - 1) Terminate metal conduit entering bottom with earthing bushing; provide an earthing jumper extending to equipment earth bus or earthing pad; and
 - 2) Terminate PVC conduit entering bottom with bell end fittings.
- (n) Underground Raceways
 - (i) Grade: Maintain minimum grade of 100 mm in 30 metres, either from one draw pit to the next, or from a high point between them, depending on surface contour;
 - (ii) Cover: Maintain minimum 600 mm cover above concrete encasement, unless otherwise shown on the Drawings;
 - (iii) Make routing changes as necessary to avoid obstructions or conflicts;
 - (iv) Couplings: In multiple conduit runs, stagger so that couplings in adjacent runs are not in same transverse line;
 - (v) Spacers
 - 1) Provide preformed, non-metallic spacers, designed for such purpose, to secure and separate parallel conduit runs in a trench or concrete encasement; and
 - 2) Install at intervals not greater than 3 metres.
 - (vi) Support conduit so as to prevent bending or displacement during backfilling or concrete placement;
 - (vii) Installation with Other Piping Systems
 - 1) Crossings: Maintain minimum 300 mm vertical separation;
 - 2) Parallel Runs: Maintain minimum 300 mm separation; and
 - 3) Installation over valves or couplings not permitted.
 - (viii) Metallic Raceway Coating: Along entire length, apply wrap-around duct band with one-half tape width overlap to obtain two complete layers;
 - (IX) Provide expansion fittings that allow minimum of 100 mm of movement in vertical conduit runs from underground where exposed conduit will be fastened to or will enter building or structure;
 - (x) Provide deflection/expansion fittings in conduit runs that exit building or structure below grade. Conduit from building wall to fitting shall be PVC-coated galvanised rigid steel;

- (xi) Concrete Encasement
 - 1) As specified in the Civil and Structural General Specifications; and
 - 2) Concrete Colour: Red.
- (xii) Backfill
 - 1) As specified in the Civil and Structural General Specifications; and
 - 2) Do not backfill until inspected by S.O..
- (o) Draw Pits and Telecom Manholes
 - (i) Excavate, shore, brace, backfill and final grade in accordance with the civil and structural general specifications;
 - (ii) Do not install until final raceway grading has been determined;
 - (iii) Install such that raceways enter at nearly right angles and as near as possible to one end of wall, unless otherwise shown on the Drawings;
 - (iv) Earthing: As specified in the general specifications;
 - (v) Install Telecom manholes in accordance to IMDA COPIF requirements as noted on Drawings; and
 - (vi) Identification: Field stamp covers with draw pit number as shown on the Drawings. Stamped numbers to be 25 mm minimum height.
- (p) Empty Raceways
 - (i) Provide permanent, removable cap over each end;
 - (ii) Provide PVC plug with pull tab for underground raceways with end bells;
 - (iii) Provide nylon pull cord; and
 - (iv) Identify, as specified in Clause Identification Devices, with waterproof tags attached to pull cord at each end and at intermediate pull point.
- (q) Identification Devices
 - (i) Raceway Tags
 - 1) Identify origin and destination;
 - 2) Install at each terminus, near midpoint, and at minimum intervals of every 20 metres of exposed raceway, whether in ceiling space or surface mounted; and
 - 3) Provide nylon strap for attachment.
 - (ii) Warning Tape: Install approximately 300 mm above underground or concrete-encased raceways. Align parallel to and within 200 mm of, centreline of runs.
 - (iii) Buried Raceway Markers
 - 1) Install at grade to indicate direction of underground raceways;
 - 2) Install at all bends and at intervals not exceeding 25 metres in straight runs; and
 - 3) Embed and secure to top of concrete base, sized 300 mm long, 150 mm wide and 200 mm deep; top set flush with finished grade.

- (r) Protection of Installed Work
 - (i) Protect products from effects of moisture, corrosion and physical damage during construction;
 - (ii) Provide and maintain manufactured watertight and dust-tight seals over all conduit openings during construction;
 - (iii) Touch up painted conduit threads after assembly to cover nicks or scars; and
 - (iv) Touch up coating damage to PVC-coated conduit with patching compound approved by manufacturer; compound shall be kept refrigerated according to manufacturer's instructions until time of use.

END OF SECTION